Imam Abdulrahman Bin Faisal University
College of Computer Science and Information Technology

# ARTI 404 — Natural Language Processing

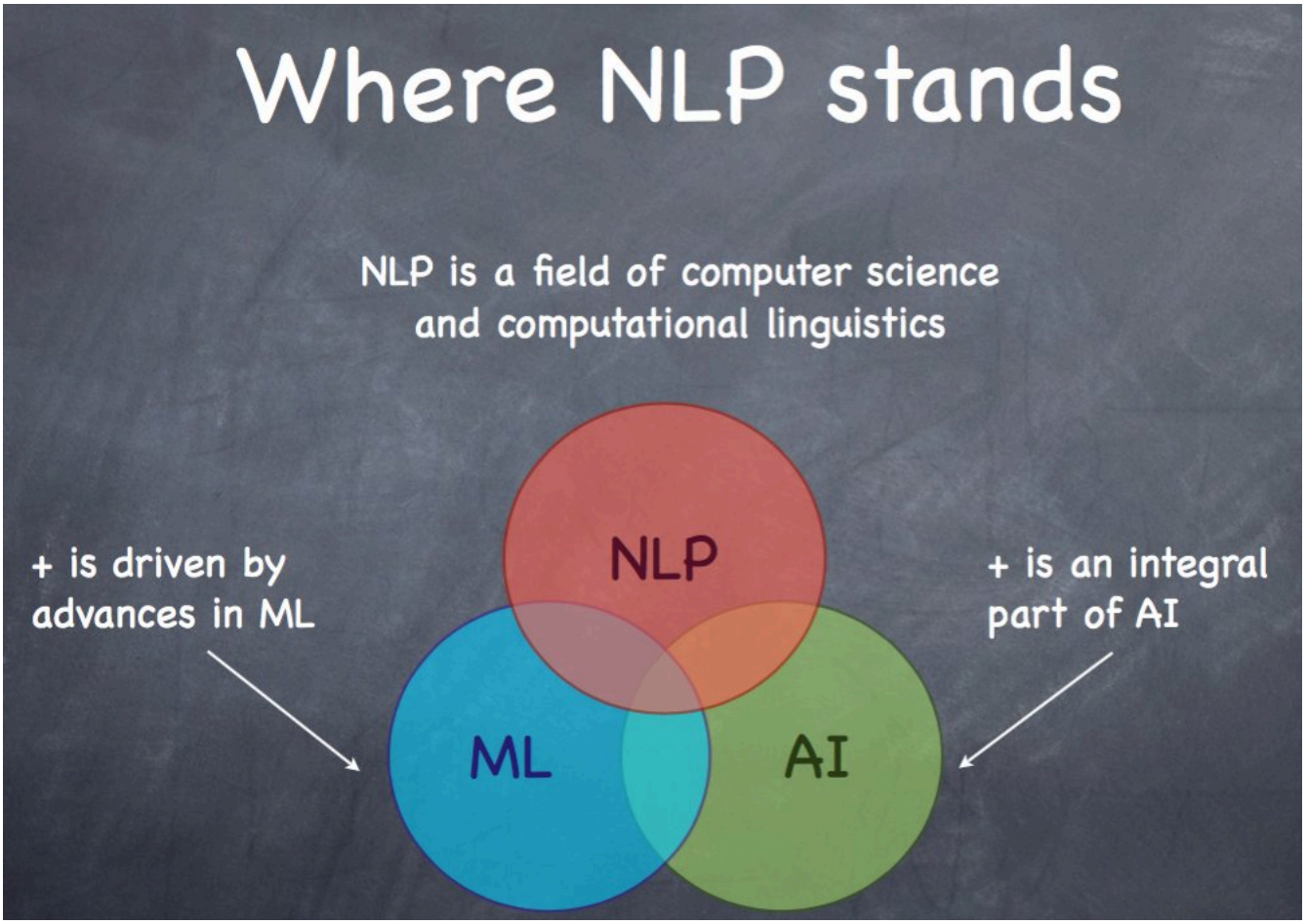
Lab 1 — Introduction to Natural Language Processing (NLP)



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# Introduction to Natural Language processing

**Natural language processing (NLP):** Subfield of linguistics, computer science, and information engineering concerned with the interactions between computers and human (natural) languages, in particular how to program computers to process and analyze large amounts of natural language data.




## What is it all about?

# What's that all about

Let's start with the basics:



we speak

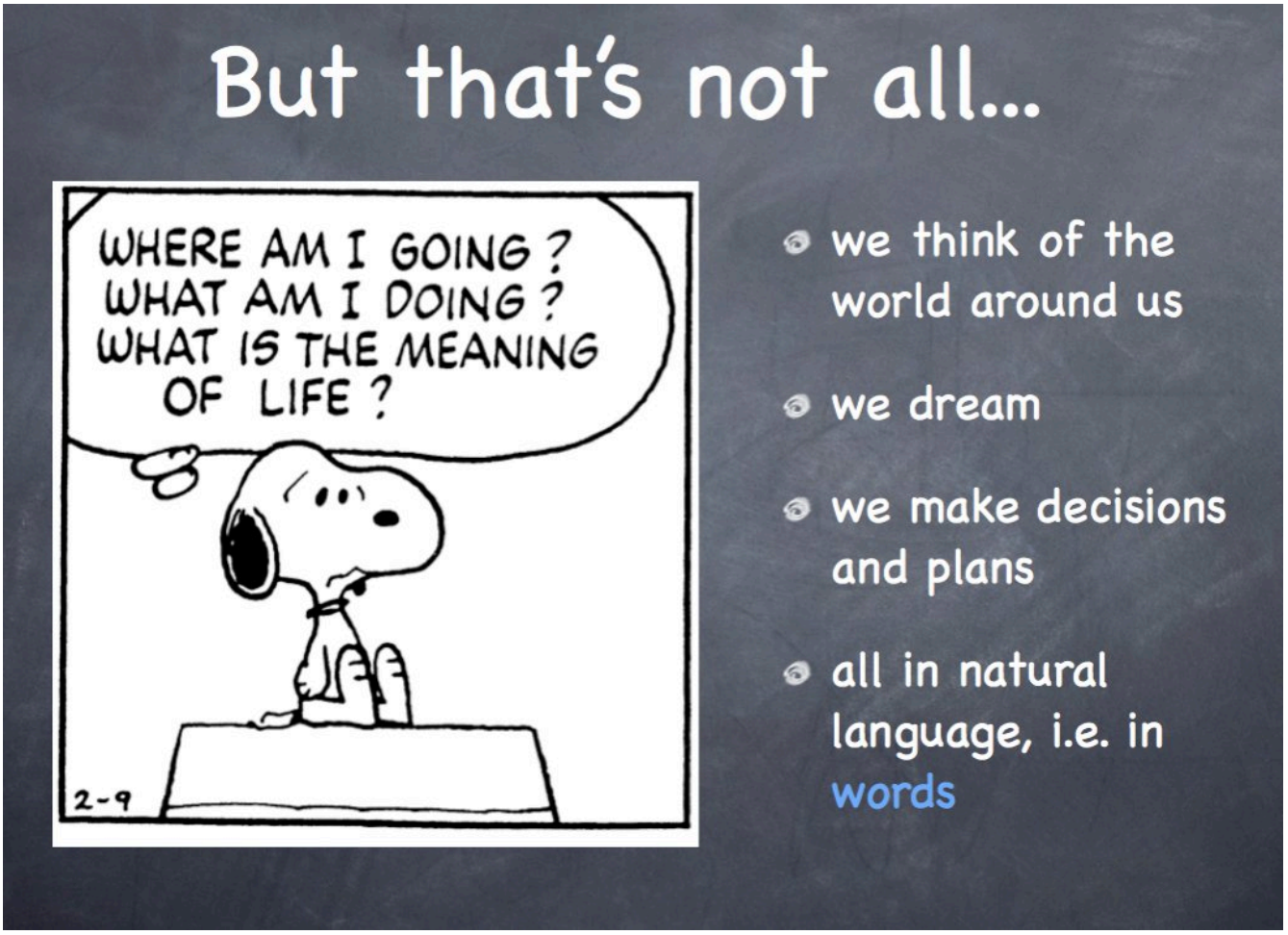


we read




we write

all that using language




## What are the raw materials for NLP revolution?

**1. Data:** There is a huge amount of human knowledge that has been digitized.

**2. Compute:** Our personal computers are very powerful, and it has never been easier to rent more compute power.

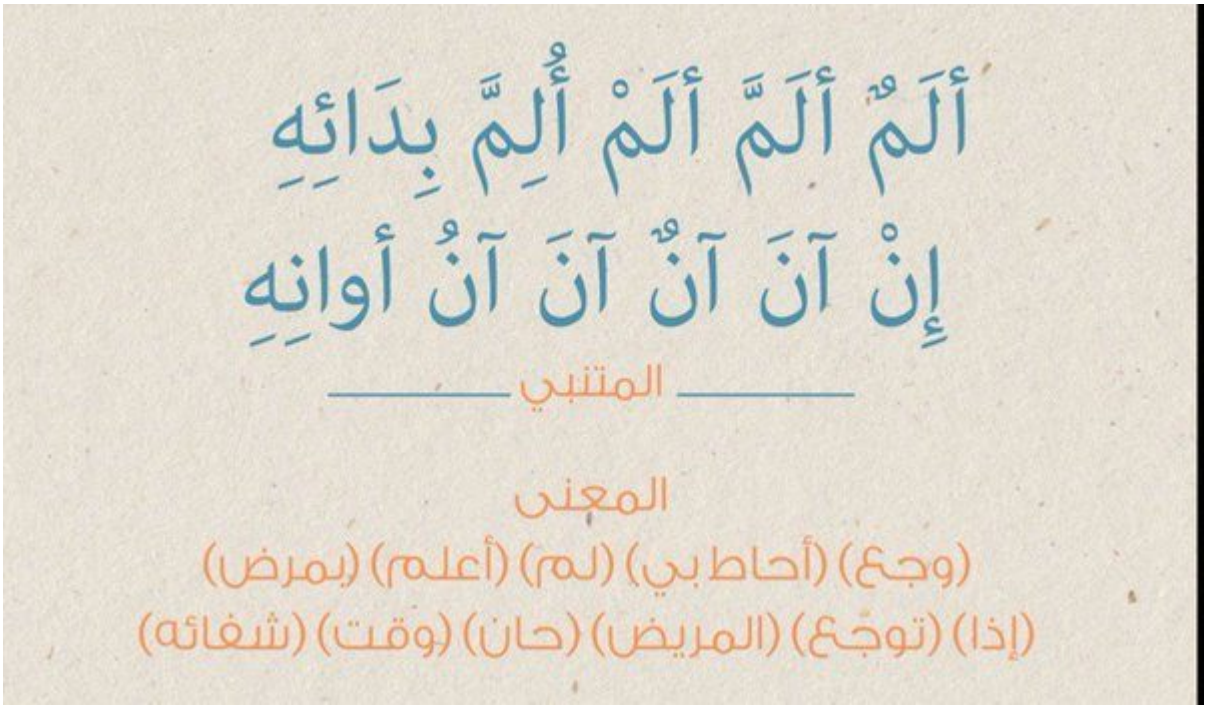
**3. Algorithms:** We use a combination of Computer Science, Machine Learning, Deep Learning, and linguist techniques.

## Why NLP is hard?

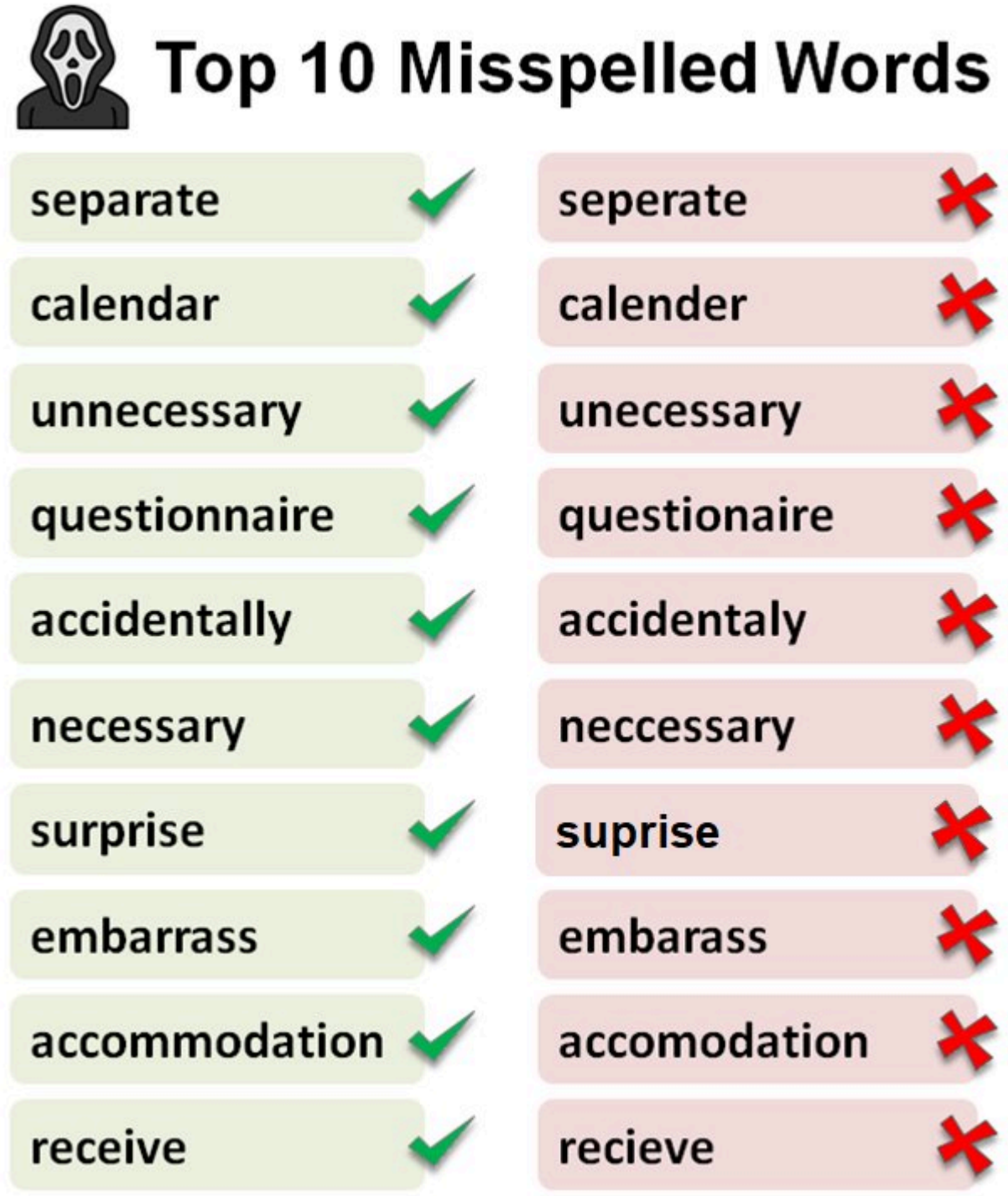
It's the nature of the human language that makes NLP difficult.

### Some examples:

**1. Ambiguity:** One word/sentence can have different meanings.




**2. Spelling Errors:** There are some words that are usually misspelled by people.




**3. Phonetics and Phonology**: Different words that have the same pronunciation.

- there / their
- plane / plain
- ice cream / I scream

**4. Order of Context:** "Dog bites man" and "Man bites dog" have vastly different meanings

### In ML terms, why is NLP hard?

**1. Input:** Large, discrete, open-ended input state spaces

**2. Output:** Variable length output spaces

**3. Algorithms:** Recycled from other fields

## Top Used Text File Types

- CSV
- Text Files
- JSON
- SQL
- HTML
- PDF
- DOCX

## NLP Libraries

There are many popular NLP libraries that we will cover in the next labs. In this lab, we will get introduced to NLTK.

### NLTK

Natural Language Toolkit (nltk) is a Python package for NLP
https://www.nltk.org/

### What are Corpus / Corpora?

a collection of written texts, especially the entire works of a particular author or a body of writing on a particular subject.

```
!pip install nltk
```

```
Requirement already satisfied: nltk in c:\users\ahmad\miniconda3\lib\site-packages (3.9.2)
Requirement already satisfied: click in c:\users\ahmad\miniconda3\lib\site-packages (from
nltk) (8.2.1)
Requirement already satisfied: joblib in c:\users\ahmad\miniconda3\lib\site-packages (from
nltk) (1.5.2)
Requirement already satisfied: regex>=2021.8.3 in c:\users\ahmad\miniconda3\lib\site-packages
(from nltk) (2025.9.18)
Requirement already satisfied: tqdm in c:\users\ahmad\miniconda3\lib\site-packages (from nltk)
(4.67.1)
Requirement already satisfied: colorama in c:\users\ahmad\miniconda3\lib\site-packages (from
click->nltk) (0.4.6)
```

```
import nltk
```

### Arabic Libraries (Extra)

Camel-Tools: https://camel-tools.readthedocs.io/en/latest/

PyArabic: https://pyarabic.readthedocs.io/ar/latest/

Farasa: https://farasa.qcri.org/

## NLP Applications

### 1. Information Extraction

The automated retrieval of specific information related to a selected topic from a body or bodies of text.

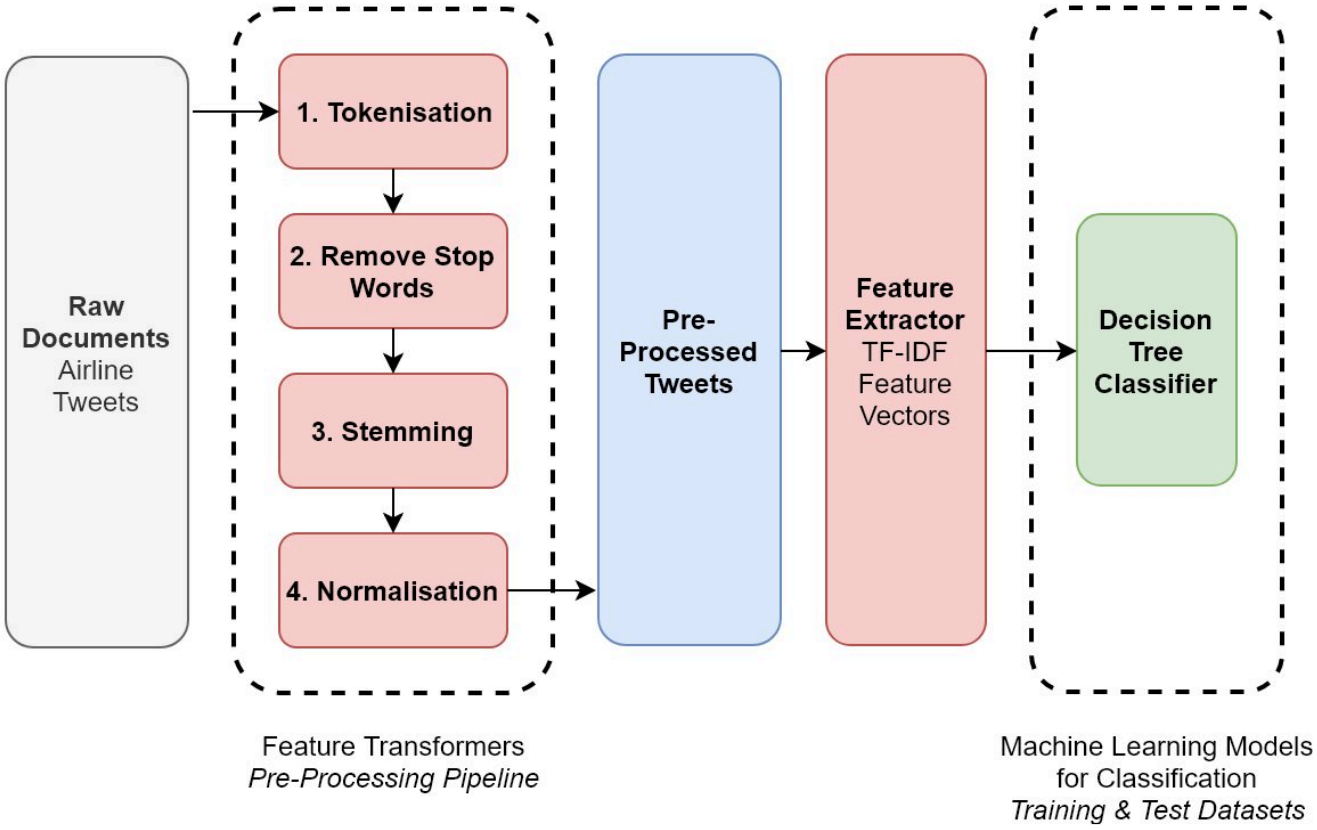
### 2. Machine Translation

The process of using artificial intelligence to automatically translate text from one language to another without human involvement.

### 3. Sentiment Analysis

Identifying sentiments and opinions stated in a text.

## Example of an NLP Pipeline




### 1. Text Acquisition:

Obtain the text data from a source, such as a file, a web page, or a database.

### 2. Text Preprocessing:

- **Tokenization:** Split the text into individual words or tokens.
- **Lowercasing:** Convert all tokens to lowercase to ensure uniformity.
- **Removing Punctuation:** Strip away punctuation marks from tokens.
- **Stopword Removal:** Eliminate common words (e.g., "and," "the," "is") that may not carry significant meaning.
- **Stemming:** Removes prefixes/suffixes to get a root form. (Running -> run)
- **Lemmatization:** Converts word to base/dictionary form (lemma). (better -> good).

### 3. Feature Extraction:

- **Bag-of-Words (BoW):** Create a vector representation of the text using the frequency of each token in the document.
- **TF-IDF (Term Frequency-Inverse Document Frequency):** Assign weights to tokens based on their importance in the document relative to the entire corpus.

### 4. Model Building:

Choose a machine learning model for sentiment analysis, such as a Naive Bayes classifier, Support Vector Machine (SVM), or a neural network.

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## Task#1:

Students should work in groups and search online for one of the different types of datasets.
• CSV
• Text Files
• JSON
• SQL
• HTML
• PDF
• DOCX

## Task#2:

Download and open the dataset.

```
conda install kagglehub
```

```
3 channel Terms of Service accepted
Channels:
 - defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

# All requested packages already installed.


Note: you may need to restart the kernel to use updated packages.
```

```
==> WARNING: A newer version of conda exists. <==
    current version: 26.1.1
    latest version: 26.7.2

Please update conda by running

    $ conda update -n base -c defaults conda
```

```
import kagglehub

# IMDB Dataset of 50K Movie Reviews
path = kagglehub.dataset_download("lakshmi25npathi/imdb-dataset-of-50k-movie-reviews")

print("Path to dataset files:", path)
```

```
Path to dataset files: C:\Users\ahmad\.cache\kagglehub\datasets\lakshmi25npathi\imdb-dataset-of-50k-movie-reviews\versions\1
```

```
import pandas as pd
import os

# load the CSV file in a dataframe (from the kagglehub download path)
df = pd.read_csv(os.path.join(path, "IMDB Dataset.csv"))

# Show the first few rows
df.head()
```

| | review | sentiment |
|---|---|---|
| **0** | One of the other reviewers has mentioned that ... | positive |
| **1** | A wonderful little production. <br /><br />The... | positive |
| **2** | I thought this was a wonderful way to spend ti... | positive |
| **3** | Basically there's a family where a little boy ... | negative |
| **4** | Petter Mattei's "Love in the Time of Money" is... | positive |

```
df.describe()
```

| | review | sentiment |
|---|---|---|
| **count** | 50000 | 50000 |
| **unique** | 49582 | 2 |
| **top** | Loved today's show!!! It was a variety and not... | positive |
| **freq** | 5 | 25000 |

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50000 entries, 0 to 49999
Data columns (total 2 columns):
 #   Column     Non-Null Count  Dtype 
---  ------     --------------  ----- 
 0   review     50000 non-null  object
 1   sentiment  50000 non-null  object
dtypes: object(2)
memory usage: 781.4+ KB
```

```python
# Check class distribution
print(df['sentiment'].value_counts())
```

```
sentiment
positive    25000
negative    25000
Name: count, dtype: int64
```

---

## Task 1 & Task 2 — Summary of the Selected Dataset

**Selected file type:** CSV

**Dataset:** IMDB Dataset of 50K Movie Reviews — obtained from Kaggle (*lakshmi25npathi/imdb-dataset-of-50k-movie-reviews*), downloaded using the `kagglehub` library and opened with `pandas.read_csv()`.

**Observations after opening the dataset:**

- The dataset contains 50,000 rows and 2 columns: `review` (the raw text of the review) and `sentiment` (the label).
- Both columns are of type *object* (text) and contain no missing values.
- The classes are perfectly balanced: 25,000 positive reviews and 25,000 negative reviews, which is useful for training a sentiment analysis model without class imbalance.
- Out of 50,000 reviews, 49,582 are unique, so a small number of reviews are duplicated.
- The reviews still contain HTML tags such as `<br />`, so text preprocessing (removing HTML tags, lowercasing, removing punctuation and stopwords, and tokenization) is required before feature extraction.

**Possible NLP task:** sentiment analysis (binary text classification of a movie review as positive or negative).